SCHEMATIC LOCUS DETECTED BY AN AFFINE OPEN CONTAINING THE FIBRE OF A FINITE LOCALLY FREE COVER

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- fppf morphisms are open. Morphisms of Spaces, lemma fppf-open (cross-chapter, named). The image of U' is open.
- Points over fibre products. Properties of Spaces, lemma points-cartesian (cross-chapter, named). U' is the inverse image of V'.
- Finite flat equivalence global. Properties of Spaces, proposition finite-flat-equivalence-global (cross-chapter, named). The fppf quotient sheaf X/R is a scheme in the reduced situation.
- Surjectivity of flat finitely presented presentations. Spaces, lemma surjective-flat-locally-finite-presentation (cross-chapter, named). X -> Y is surjective as a map of sheaves.

Source reference. Original reference 05VM: focused-seeds/geometry/S9-decent-when-quotient-scheme-at-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 042S: focused-seeds/geometry/S9-decent-when-quotient-scheme-at-point/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-when-quotient-scheme-at-point/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07S6: focused-seeds/geometry/S9-decent-when-quotient-scheme-at-point/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be a surjective finite locally free morphism of algebraic spaces over S. For $y \in |Y|$ the following are equivalent*

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *y is in the schematic locus of Y, and*
(2) *there exists an affine open $U \subset X$ containing the preimage of y.*

Proof. If $y \in Y$ is in the schematic locus, then it has an affine open neighbourhood $V \subset Y$ and the inverse image U of V in X is an open finite over V, hence affine. Thus (1) implies (2).

Conversely, assume that $U \subset X$ as in (2) is given. Set $R = X \times_Y X$ and denote the projections $s, t : R \to X$. Consider $Z = R \setminus s^{-1}(U) \cap t^{-1}(U)$. This is a closed subset of R. The image $t(Z)$ is a closed subset of X which can loosely be described as the set of points of X which are R-equivalent to a point of $X \setminus U$. Hence $U' = X \setminus t(Z)$ is an R-invariant, open subspace of X contained in U which contains the fibre of $X \to Y$ over y. Since $X \to Y$ is open (Morphisms of Spaces, Lemma 042S) the image of U' is an open subspace $V' \subset Y$. Since U' is R-invariant and $R = X \times_Y X$, we see that U' is the inverse image of V' (use Properties of Spaces, Lemma 03H4). After replacing Y by V' and X by U' we see that we may assume X is a scheme isomorphic to an open subscheme of an affine scheme.

Assume X is a scheme isomorphic to an open subscheme of an affine scheme. In this case the fppf quotient sheaf X/R is a scheme, see Properties of Spaces, Proposition 07S6. Since Y is a sheaf in the fppf topology, obtain a canonical map $X/R \to Y$ factoring $X \to Y$. Since $X \to Y$ is surjective finite locally free, it is surjective as a map of sheaves (Spaces, Lemma 05VM). We conclude that $X/R \to Y$ is surjective as a map of sheaves. On the other hand, since $R = X \times_Y X$ as sheaves we conclude that $X/R \to Y$ is injective as a map of sheaves. Hence $X/R \to Y$ is an isomorphism and we see that Y is representable. □